

COGNIFYZ TECHNOLOGIES

DATA SCIENCE INTERNSHIP

LEVEL 1 REPORT

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Domain: Data Science

Tools Used: Python, Pandas, Matplotlib, Jupyter Notebook, Anaconda

Introduction

The objective of this project is to perform data exploration, descriptive analysis, and geospatial analysis on the restaurant dataset provided by Cognifyz Technologies. Various data analysis techniques were applied to understand the dataset and extract meaningful insights.

Task 1: Data Exploration and Preprocessing

Objective

To explore the dataset, identify missing values, perform preprocessing, and analyze the target variable.

Dataset Information

- Number of Rows: 9551
- Number of Columns: 21

Missing Values Analysis

The dataset contained missing values only in the "Cuisines" column (9 records). These missing values were handled by replacing them with the value "Unknown".

Aggregate Rating Analysis

- Mean Rating: 2.67
- Median Rating: 3.20
- Standard Deviation: 1.52

- Minimum Rating: 0.00
- Maximum Rating: 4.90

Observations

- The dataset contains restaurant information from multiple cities and countries.
- Only one column contained missing values.
- Aggregate Rating shows an uneven distribution.
- Many restaurants have ratings close to zero, indicating class imbalance.

Screenshots

(Insert Dataset Head Screenshot)

(Insert Dataset Shape Screenshot)

(Insert Missing Values Screenshot)

(Insert Aggregate Rating Histogram Screenshot)

Task 2: Descriptive Analysis

Objective

To calculate statistical measures and analyze categorical variables such as city and cuisines.

Findings

- North Indian cuisine is the most common cuisine.
- Chinese cuisine is the second most common cuisine.
- New Delhi contains the highest number of restaurants.
- Fast Food, Mughlai, and Cafe cuisines are also highly represented.

Observations

- Restaurant distribution is concentrated in specific cities.
- Customer food preferences are dominated by North Indian cuisine.
- Multiple cuisine categories are represented in the dataset.

Screenshots

(Insert Descriptive Statistics Screenshot)

(Insert Top Cities Chart Screenshot)

(Insert Top Cuisines Chart Screenshot)

Task 3: Geospatial Analysis

Objective

To analyze restaurant locations using latitude and longitude coordinates.

Findings

Restaurant locations were visualized using scatter plots. The geographical distribution of restaurants was examined, and the relationship between location and ratings was analyzed.

Observations

- Restaurants are distributed across different geographical regions.
- Certain regions have a higher concentration of restaurants.
- No strong correlation was observed between geographical location and restaurant ratings.

Screenshots

(Insert Restaurant Location Scatter Plot)

(Insert City Distribution Chart)

(Insert Latitude vs Aggregate Rating Plot)

Conclusion

The restaurant dataset was successfully explored, cleaned, and analyzed. Missing values were handled appropriately, descriptive statistics were generated, and geospatial patterns were identified. The project provided valuable insights into restaurant ratings, cuisines, customer preferences, and geographical distribution.

The objectives of all Level 1 tasks were successfully completed.